

A120591 concise calculus certificate

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Definition and first terms

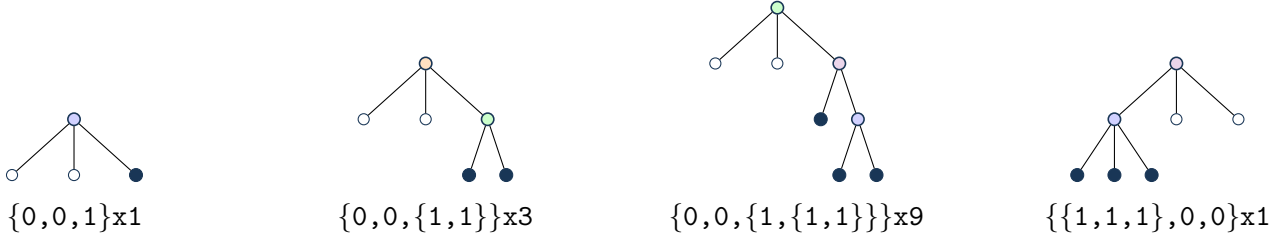
$A(x) = A_{\text{parent}}(x)^3$, $A_{\text{parent}}(x) = A_{120590}(x)$, with the origin-normalized solution.

Normalization/grammar: ordered forest of 3 parent A120590 typogeometries.

$a(0), \dots, a(7) = 1, 3, 12, 76, 600, 5304, 50232, 498360$.

Kernel used throughout: $\rho(u) = u(1 - 3u^1 - 1u^2)$, so $x = \rho(T(x))$.

Typogeometry



forest slots are explicit; 0 denotes an empty slot. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2: 3$, $\Delta_3: 1$.

Coefficient and contour extraction

$$p(n) = [x^n] A_{\text{parent}}(x),$$

$$a(n) = \sum_{\substack{n_1, \dots, n_3 \geq 0 \\ n_1 + \dots + n_3 = n}} \prod_{j=1}^3 p(n_j).$$

$$[x^n] A(x)^3 = \frac{3}{2\pi i n} \oint_{\gamma} \frac{(1+u)^2 du}{u^n D(u)^n}.$$

Recurrence

$\sum_{r=0}^3 P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$P_0(n) = 0$$

$$P_1(n) = -27n^2 - 54n - 24$$

$$P_2(n) = -162n^2 - 567n - 486$$

$$P_3(n) = 13n^2 + 65n + 78$$

Differential equation

$$(3x^2) A(x) + (-27x^3 - 81x^2) A'(x) + (-27x^4 - 162x^3 + 13x^2) A^{(2)}(x) = 24x^3 + 72x^2.$$

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n, u) = N(n, u) / (-u^3 - 3u^2 + u)^2.$$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 6×6 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.